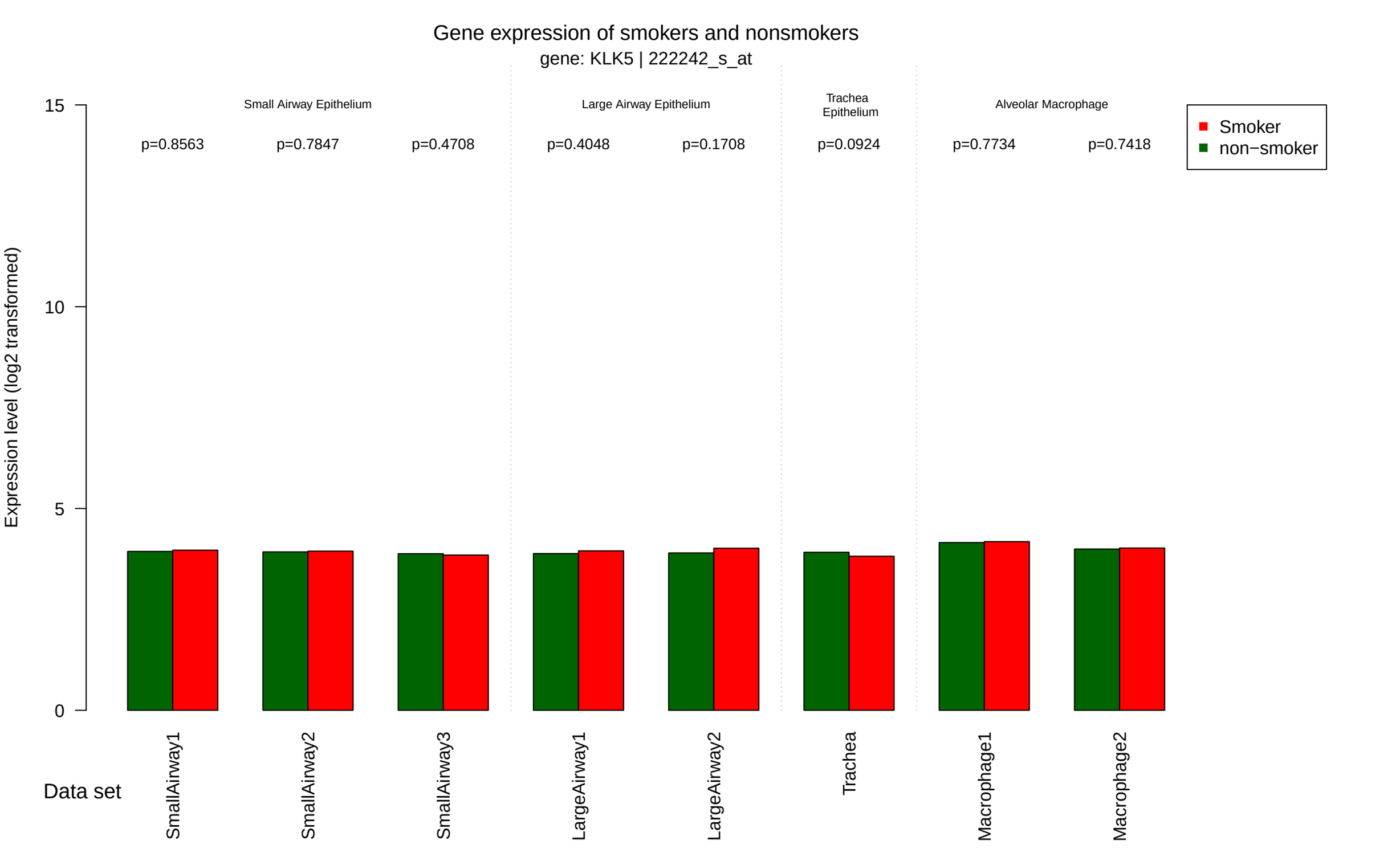
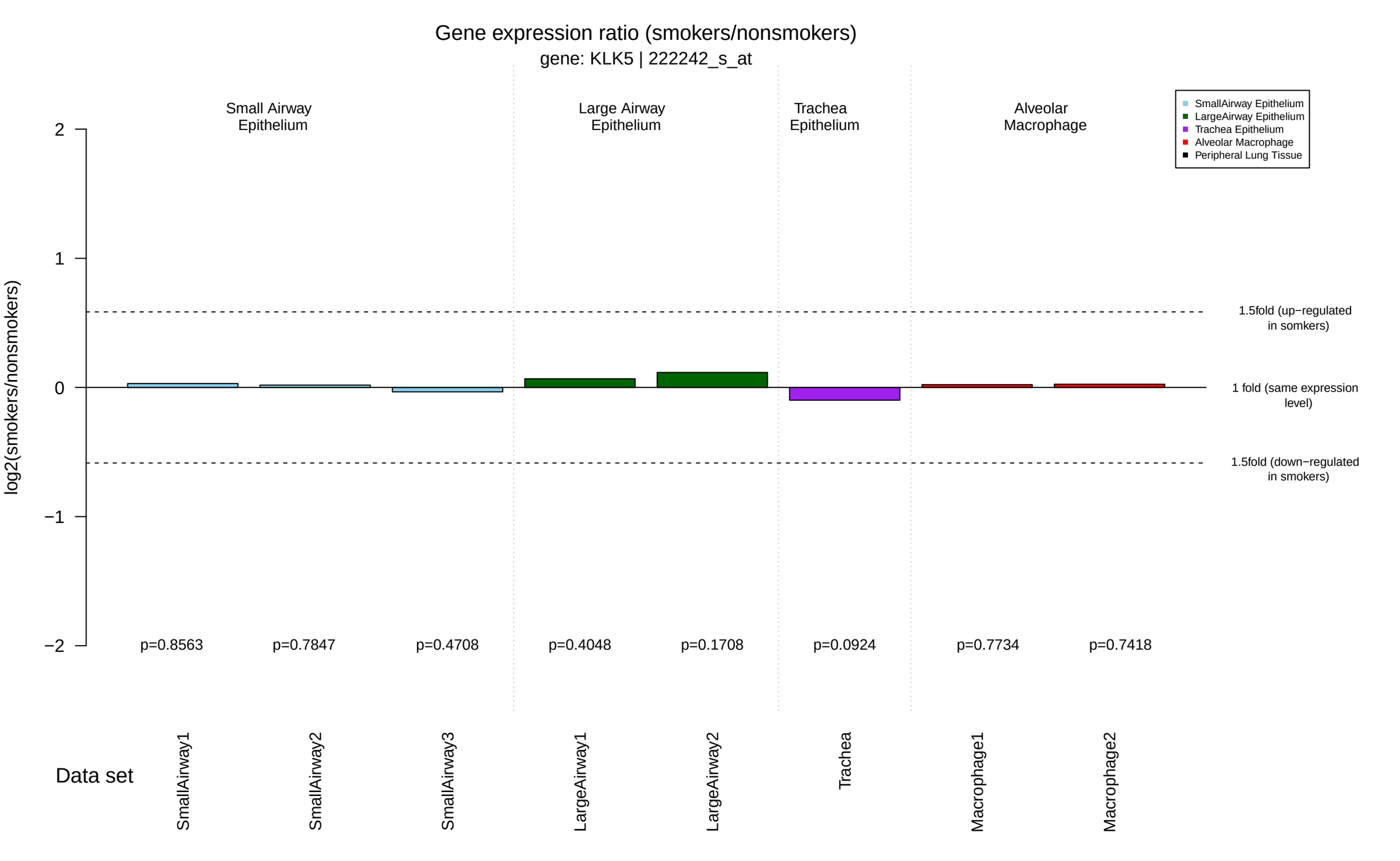
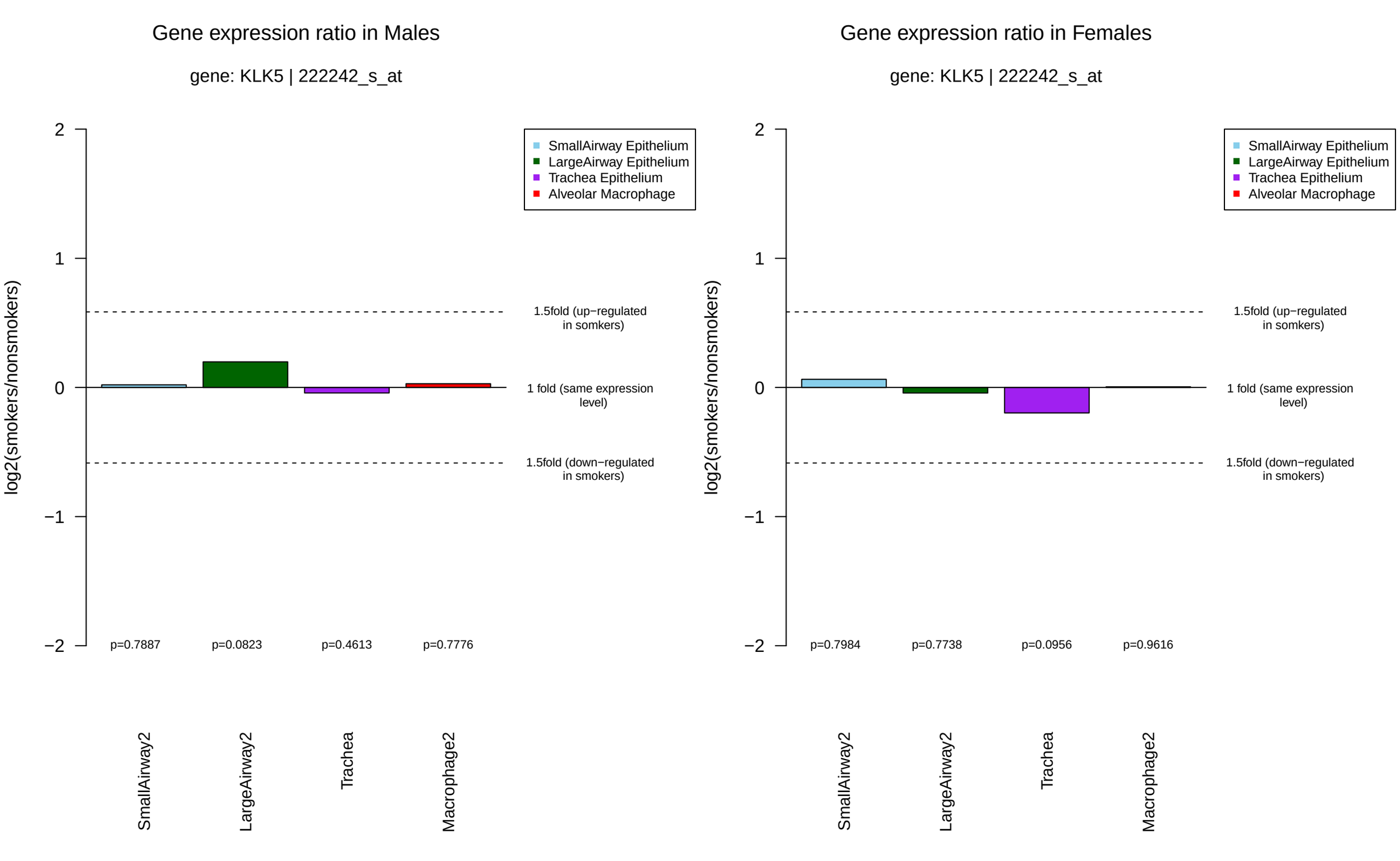
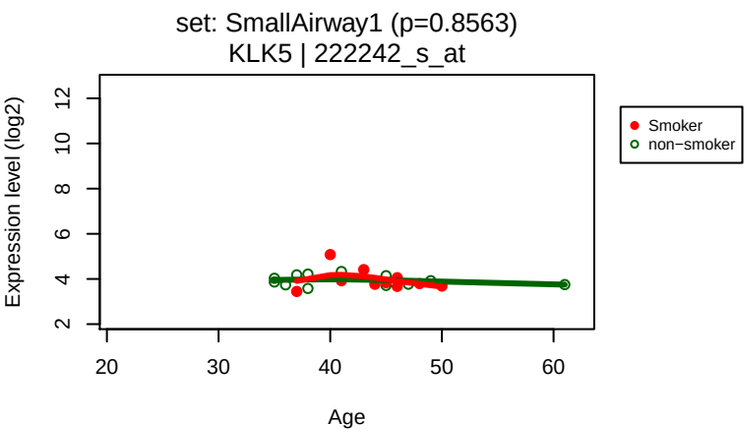
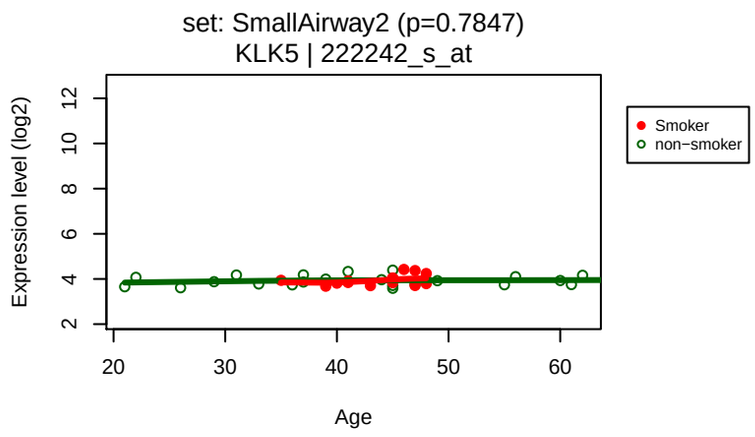
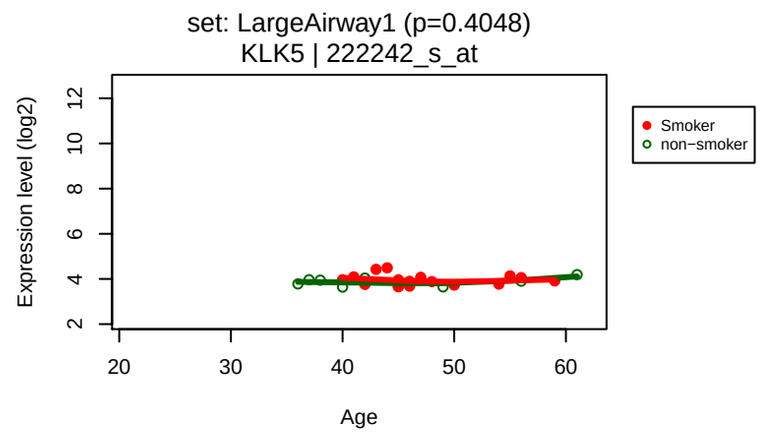

Gene expression of smokers and nonsmokers
gene: KLK5 | 222242_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8563
p=0.7847
p=0.4708
p=0.4048
p=0.1708
p=0.0924
p=0.7734
p=0.7418
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

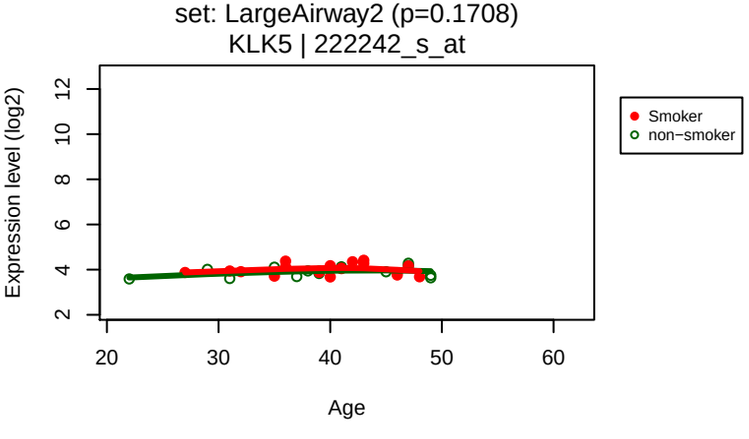

Gene expression ratio (smokers/nonsmokers)
gene: KLK5 | 222242_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8563
p=0.7847
p=0.4708
p=0.4048
p=0.1708
p=0.0924
p=0.7734
p=0.7418
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

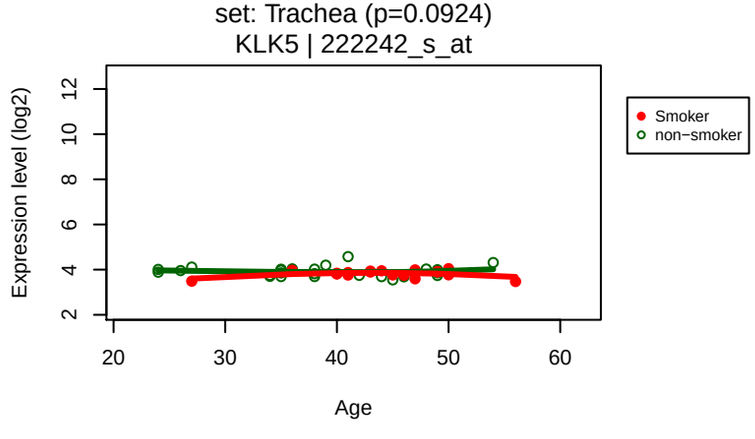

Gene expression ratio in Males
gene: KLK5 | 222242_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7887
p=0.0823
p=0.4613
p=0.7776
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK5 | 222242_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7984
p=0.7738
p=0.0956
p=0.9616
SmallAirway2
LargeAirway2
Trachea
Macrophage2

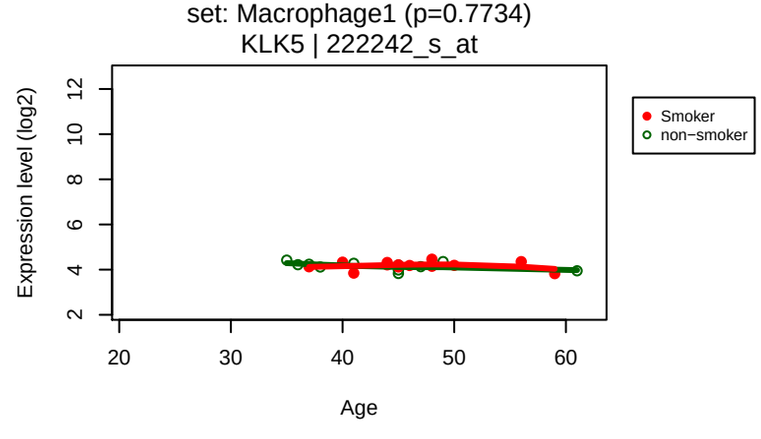

set: SmallAirway1 (p=0.8563)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker

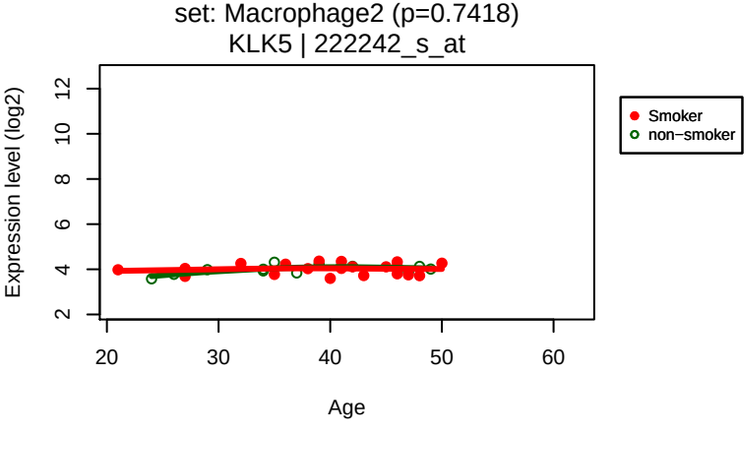

set: SmallAirway2 (p=0.7847)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker

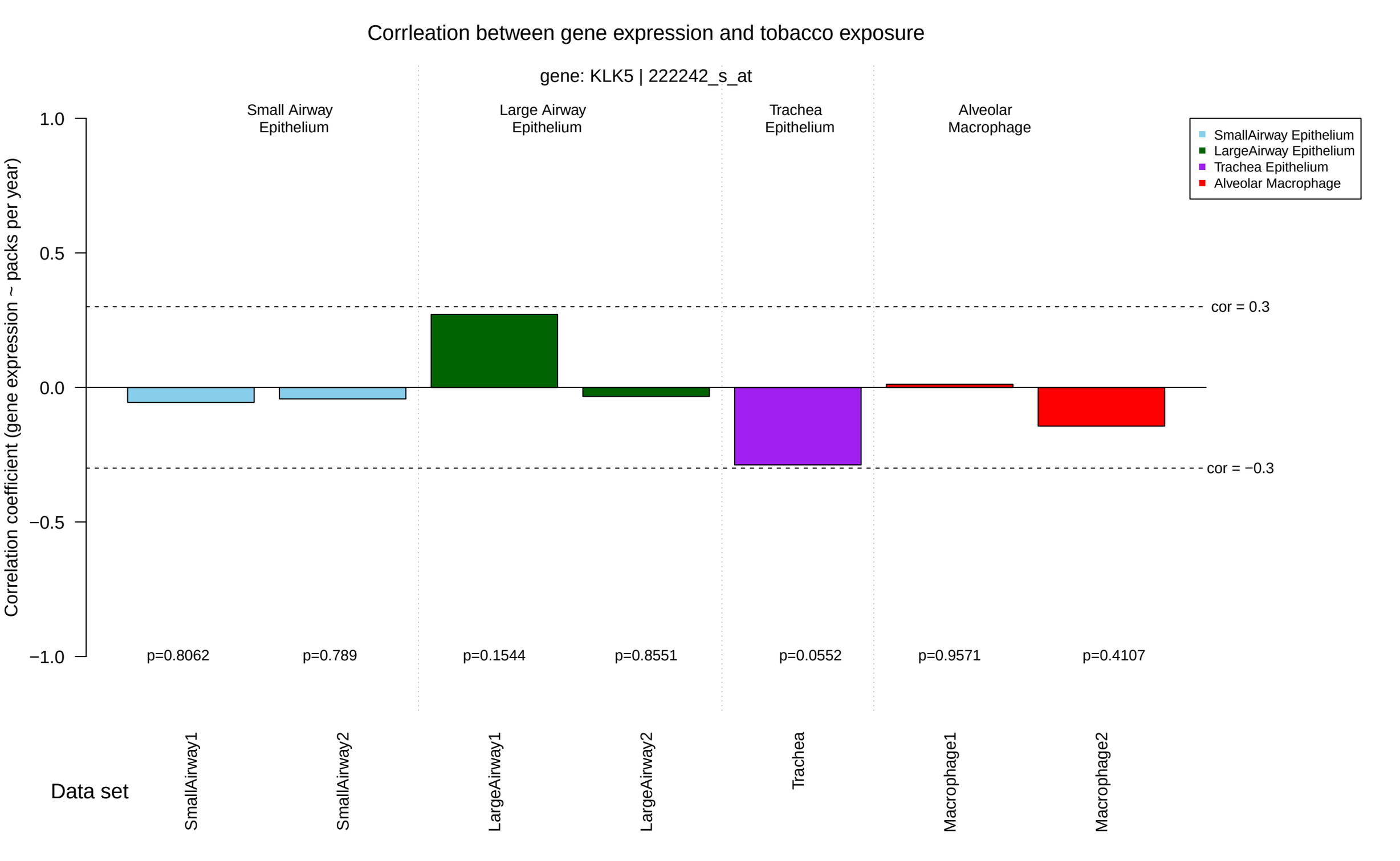

set: LargeAirway1 (p=0.4048)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1708)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0924)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7734)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7418)
KLK5 | 222242_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK5 | 222242_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8062
p=0.789
p=0.1544
p=0.8551
p=0.0552
p=0.9571
p=0.4107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2